Uncertainty intervals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete interval-input, assumption and interpretation record.

Keep target, mean/sigma/n, formula, units, bounds, method status and evidence limits together.

Keep one habit: Recall → W1–W3/W5 → independent interval calculation → uncertainty project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Sampling variability

> **My remembered idea, example and correction**

Experimental units and independence

> **My remembered idea, example and correction**

A literature comparison matrix

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An uncertainty interval needs a target, sampling model and assumptions. Main exercise: independent normal measurements and known positive population sigma. With mean 10 units, sigma 2 units and n=16, sqrt(n)=4, SE=0.5, margin=1.96×0.5=0.98, and interval [9.02, 10.98]. Rounded critical value 1.96 gives approximately 95% theoretical repeated-sampling coverage.

  This interval concerns the population mean; it is not a range for 95% of individual values or a posterior probability attached to the fixed mean after observation. Unknown sigma requires a justified alternative method, and biased or dependent sampling is not repaired by arithmetic. Separate picture supplies B–A estimate −0.8 seconds and appropriate-method 95% bounds [−1.4, −0.2], entirely negative.

  Changed supplied bounds [−1.4, 0.3] include both signs and zero, not proof of equality. Raw data and fitting details are absent: no SE, symmetric reconstruction or refitting claimed. Neither example establishes every-request speed, causality or practical value. Code validates finite nondegenerate arithmetic, not real sampling assumptions.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- An uncertainty interval needs a target, sampling model and assumptions. Main exercise: independent normal measurements and known positive population sigma. With mean 10 units, sigma 2 units and n=16, sqrt(n)=4, SE=0.5, margin=1.96×0.5=0.98, and interval [9.02, 10.98]. Rounded critical value 1.96 gives approximately 95% theoretical repeated-sampling coverage.

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- Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

Main calculation step	Substitution	Result and unit
Root sample size	sqrt(16)	4, dimensionless
Standard error	sigma / sqrt(n) =2/4	0.5 measurement units

Main calculation step	Substitution	Result and unit
Margin	1.96 ×0.5	0.98 measurement units
Lower endpoint	10 −0.98	9.02 measurement units
Upper endpoint	10 +0.98	10.98 measurement units
Reported interval	[9.02, 10.98]	For the population mean under stated assumptions
What is not calculated	Raw individual-value coverage or unknown-sigma interval	No actual sampling or coverage experiment

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An uncertainty interval needs a target, sampling model and assumptions. Main exercise: independent normal measurements and known positive population sigma. With mean 10 units, sigma 2 units and n=16, sqrt(n)=4, SE=0.5, margin=1.96×0.5=0.98, and interval [9.02, 10.98]. Rounded critical value 1.96 gives approximately 95% theoretical repeated-sampling coverage.

  This interval concerns the population mean; it is not a range for 95% of individual values or a posterior probability attached to the fixed mean after observation. Unknown sigma requires a justified alternative method, and biased or dependent sampling is not repaired by arithmetic. Separate picture supplies B–A estimate −0.8 seconds and appropriate-method 95% bounds [−1.4, −0.2], entirely negative.

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- A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- An uncertainty interval needs a target, sampling model and assumptions. Main exercise: independent normal measurements and known positive population sigma. With mean 10 units, sigma 2 units and n=16, sqrt(n)=4, SE=0.5, margin=1.96×0.5=0.98, and interval [9.02, 10.98]. Rounded critical value 1.96 gives approximately 95% theoretical repeated-sampling coverage.

  This interval concerns the population mean; it is not a range for 95% of individual values or a posterior probability attached to the fixed mean after observation. Unknown sigma requires a justified alternative method, and biased or dependent sampling is not repaired by arithmetic. Separate picture supplies B–A estimate −0.8 seconds and appropriate-method 95% bounds [−1.4, −0.2], entirely negative.

  Changed supplied bounds [−1.4, 0.3] include both signs and zero, not proof of equality. Raw data and fitting details are absent: no SE, symmetric reconstruction or refitting claimed. Neither example establishes every-request speed, causality or practical value. Code validates finite nondegenerate arithmetic, not real sampling assumptions.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled uncertainty intervals **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of uncertainty intervals.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Uncertainty intervals. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Uncertainty intervals. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

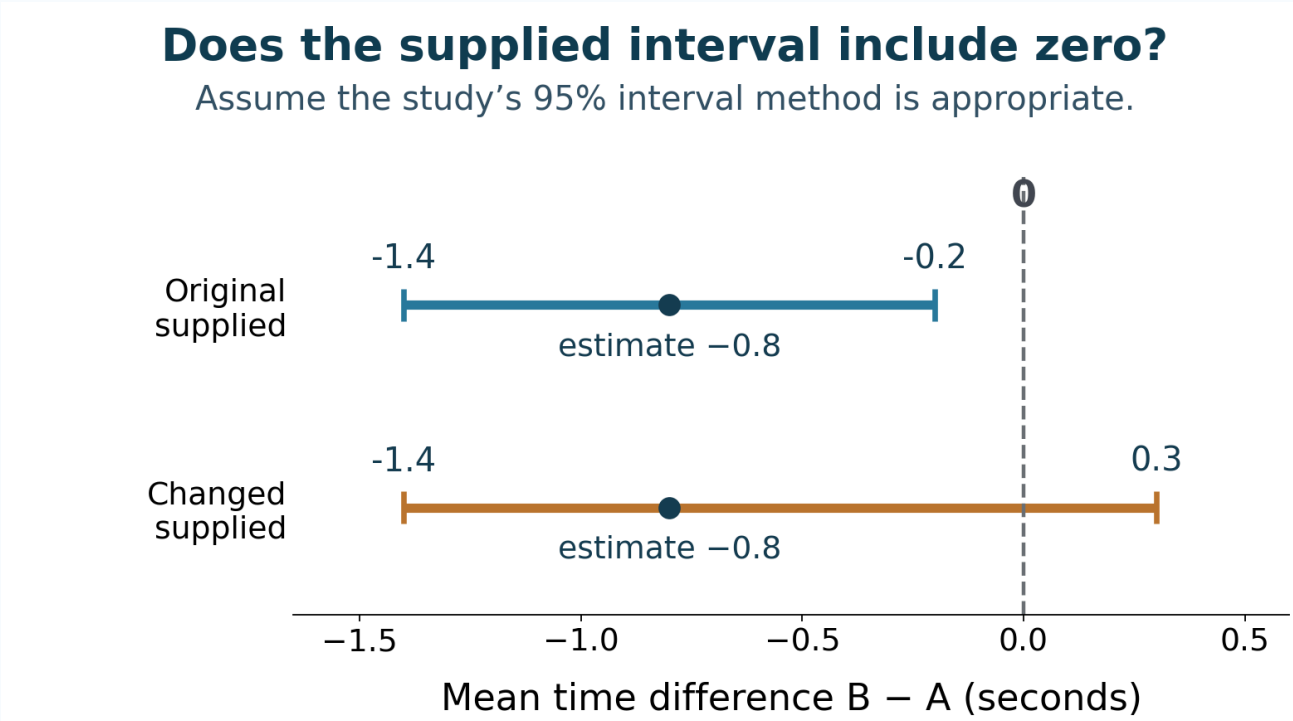
- Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

- Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

- Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/12

Supplied interval for B minus A time

	Lower endpoint (s)	Estimate (s)	Upper endpoint (s)
Original supplied interval	-1.4	-0.8	-0.2
Changed supplied interval	-1.4	not specified	0.3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/12

Uncertainty intervals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate the estimate

-0.8 lies between the interval endpoints.



2. Interpret the sign

Negative B-A means B took less time under this measurement convention.



3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

CHANGE ONE THING

Use a supplied interval [-1.4,0.3] instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.
- Which labelled step matches this detail? 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.
- 1. Interpret the sign
- 2. Locate the estimate
- 3. Respect coverage meaning
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Respect coverage meaning 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Changed-case question and optional hint

- Changed-case practice: Use a supplied interval [-1.4,0.3] instead. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Repair and retain the record

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the **mechanism** and terminology of a literature comparison matrix.

Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Uncertainty intervals: the case

AI ACADEMY / AI-7/12 · 01 · LOOK AT THE FACTS

Uncertainty intervals: the case

The situation

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Supplied evidence

Estimated difference B−A is -0.8 seconds; supplied 95% confidence interval is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

Explain uncertainty intervals using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

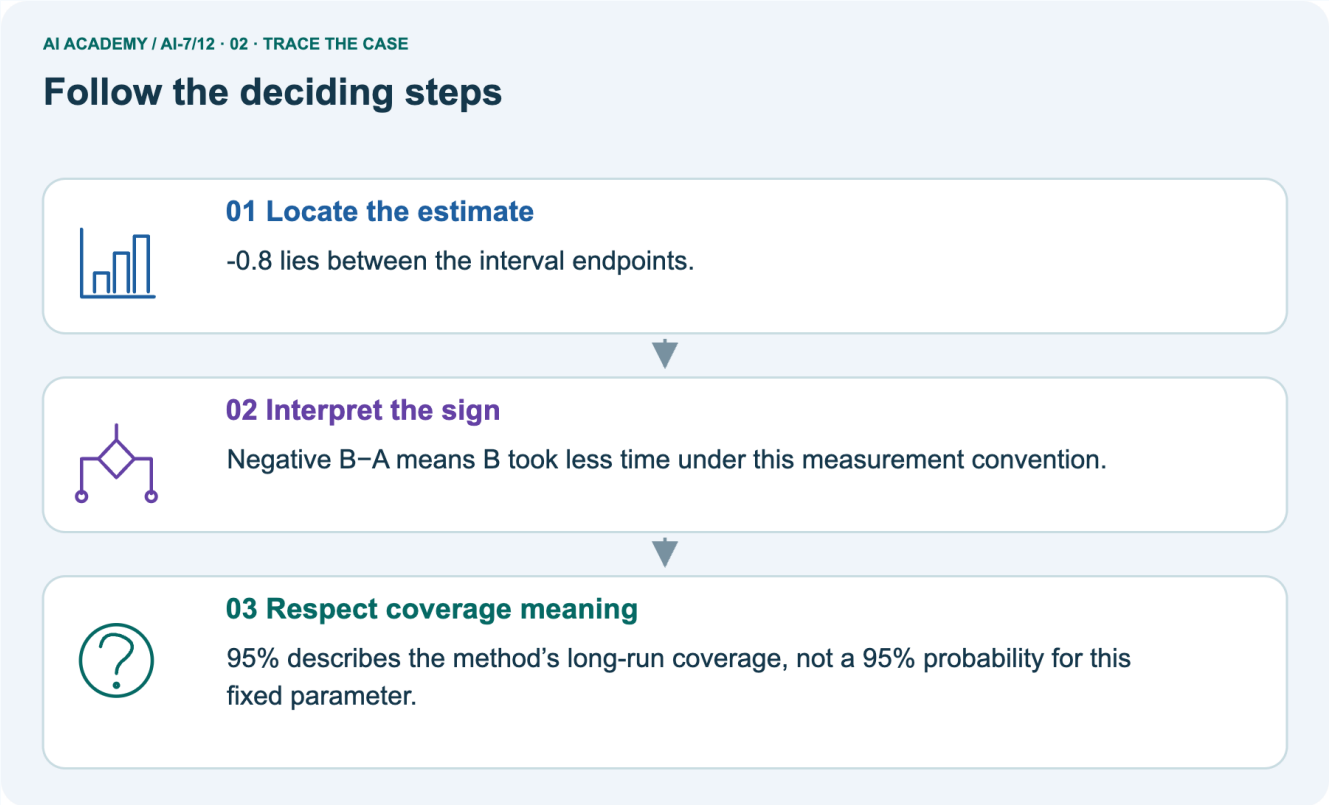
Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Supplied facts: Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Locate the estimate: -0.8 lies between the interval endpoints. → Interpret the sign: Negative B–A means B took less time under this measurement convention. → Respect coverage meaning: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Supplied interval for B minus A time



Follow the labelled picture with the matching explanation. Open the vector diagram

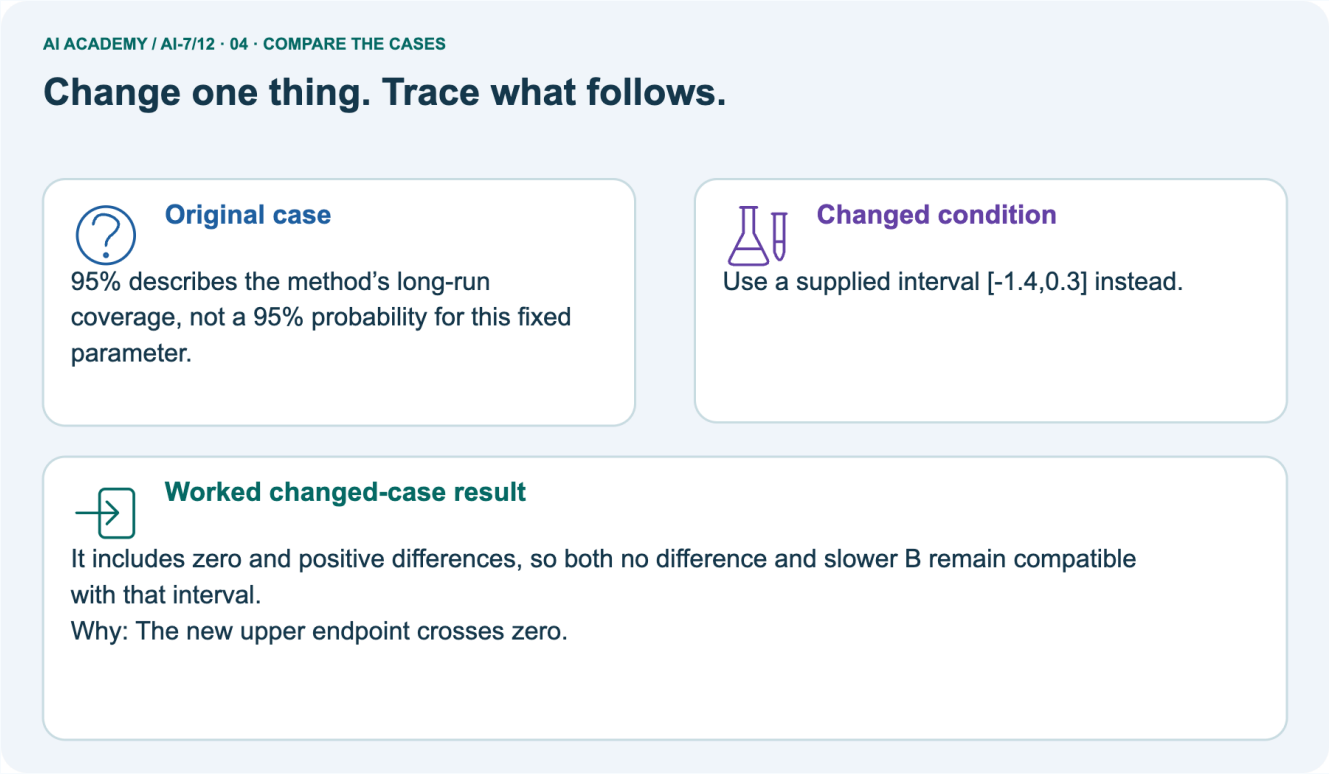
Supplied interval for B minus A time. Original supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = -0.8, Upper endpoint (s) = -0.2.

Changed supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = not specified, Upper endpoint (s) = 0.3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

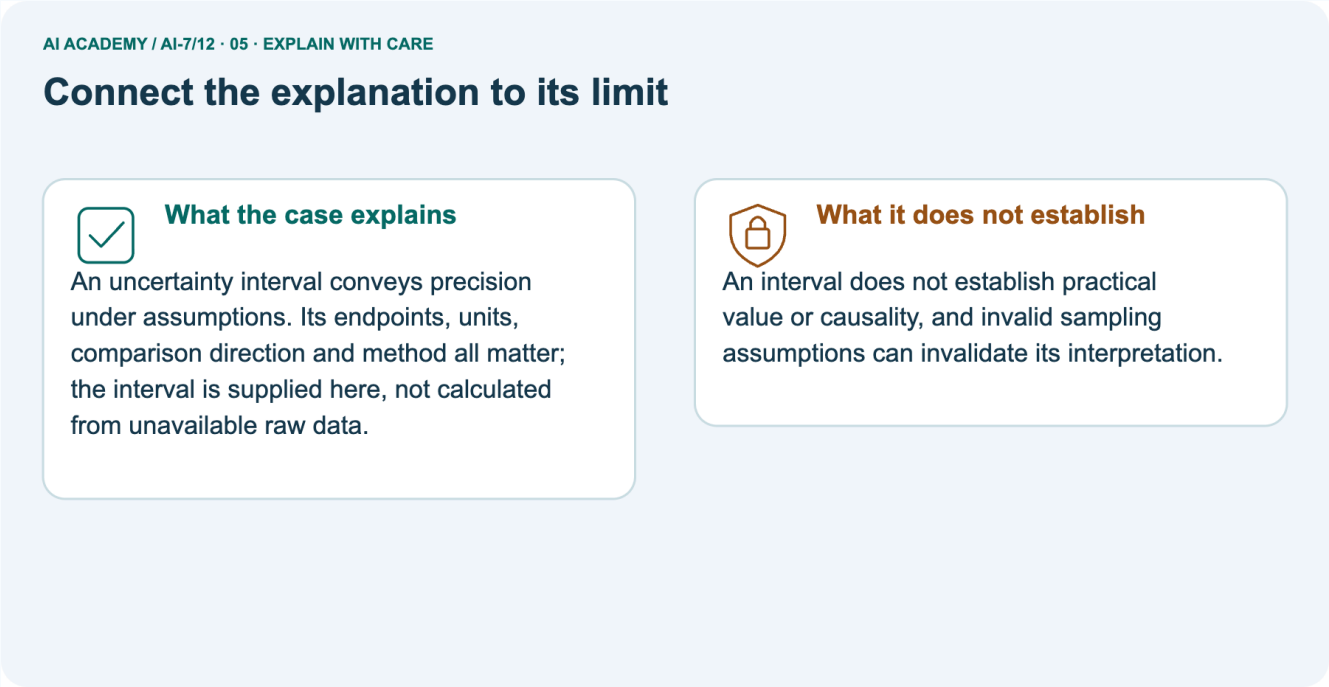
Original case: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter. Changed condition: Use a supplied interval [-1.4,0.3] instead.

Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

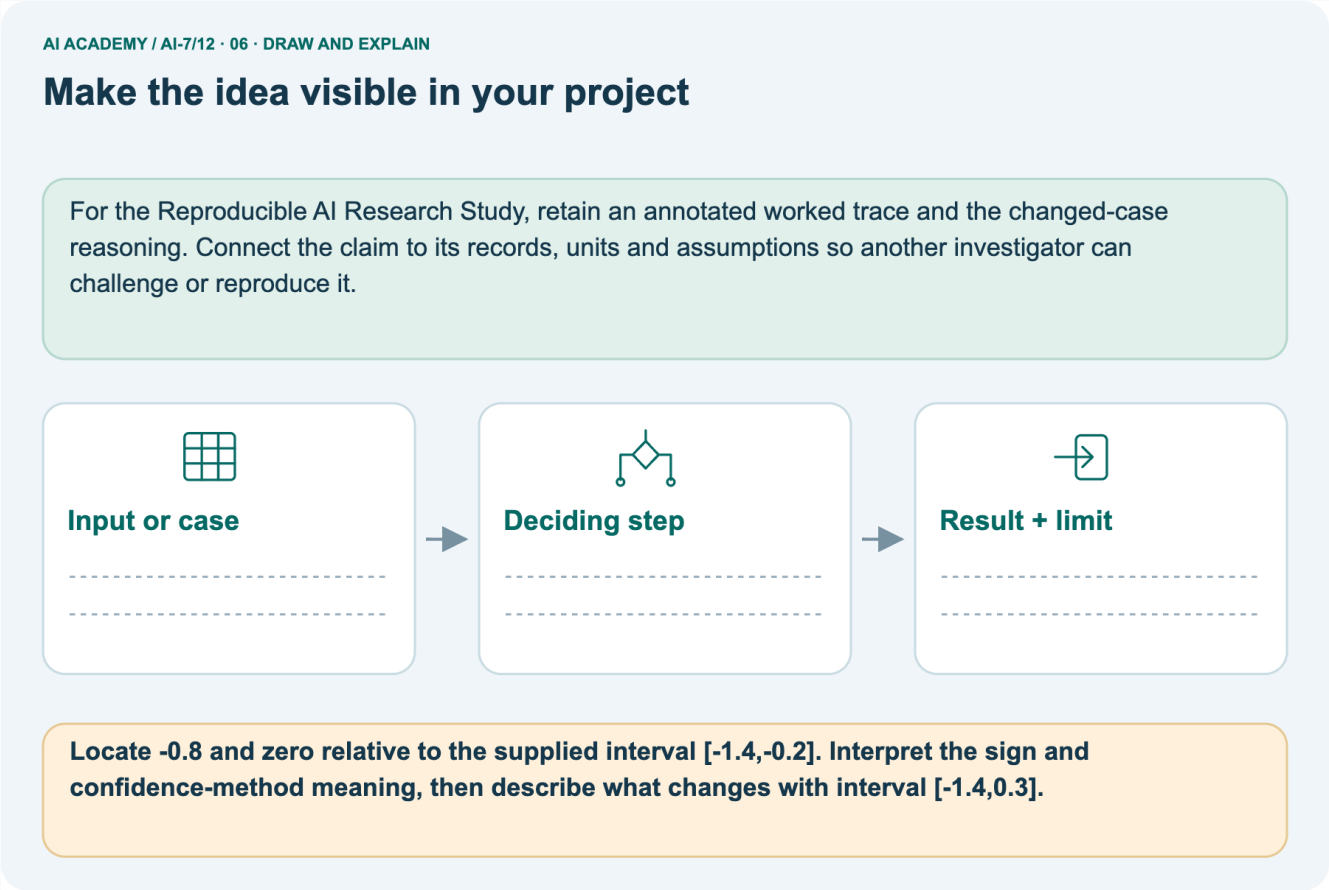
An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Limit: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Uncertainty intervals: the case

02. Follow the deciding steps

03. Supplied interval for B minus A time

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.